

assignment kine1D free_fall

 This is a preview of the published version of the quiz

Started: Apr 29 at 7:28pm

Quiz Instructions

2 attempts



Question 1 12.5 pts

A stone is thrown with an initial upward velocity of 7.0 m/s and experiences negligible air resistance. If we take upward as the positive direction, what is the velocity of the stone after 0.50 s?

☐

2.1 m/s

☐

-4.9 m/s

☐

0.00 m/s

☐

4.9 m/s

☐

-2.1 m/s



Question 2 12.5 pts

A ball is thrown straight up with a speed of 36 m/s. How long does it take to return to its starting point, assuming negligible air resistance?

☐

3.7 s

☐

7.3 s

☐

11 s

☐

15 s



Question 3 12.5 pts

A laser is thrown upward with a speed of 12 m/s on the surface of planet X where the acceleration due to gravity is 1.5 m/s^2 and there is no atmosphere. What is the maximum height reached by the laser? (hint: $a = -1.5 \text{ m/s/s}$)

☐

18 m

☐

8.0 m

☐

48 m

☐

144 m



Question 4 12.5 pts

A bullet shot straight up returns to its starting point in 10 s. What is the initial speed of the bullet, assuming negligible air resistance?

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9.8 m/s

☐

25 m/s

☐

49 m/s

☐

98 m/s



Question 5 12.5 pts

A ball is projected upward at time $t = 0$ s, from a point on a flat roof 10 m above the ground. The ball rises and then falls with insignificant air resistance, missing the roof, and strikes the ground. The initial velocity of the ball is 58.5 m/s. Consider all quantities as positive in the upward direction. At time $t = 5.97$ s the vertical velocity of the ball is closest to

☐

-175 m/s.

☐

+175 m/s.

☐

0 m/s.

☐

-12 m/s.

☐

+12 m/s.



Question 6 12.5 pts

A ball is thrown downward from the top of a building with an initial speed of 25 m/s. It strikes the ground after 2.0 s. How high is the building, assuming negligible air resistance?

☐

30 m

☐

50 m

☐

20 m

☐

70 m



Question 7 12.5 pts

A laser is thrown upward with a speed of 12 m/s on the surface of planet X where the acceleration due to gravity is 1.5 m/s^2 and there is no atmosphere. How long does it take for the laser to reach the maximum height?

☐

8.0 s

☐

11 s

☐

14 s

☐

16 s



Question 8 12.5 pts

Human reaction time is usually greater than 0.10 s. If your friend holds a ruler between your fingers and releases it without warning, how far can you expect the ruler to fall before you catch it, assuming negligible air resistance?

☐

at least 9.8 cm

☐

at least 4.9 cm

☐

at least 3.0 cm

☐

at least 6.8 cm

Not saved

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